Capacity-Law Serving of a 235B Mixture-of-Experts Model on 32 GB Host Memory

Routing Traces and Union Demand on a Consumer SSD

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Scope. This manuscript is the measurement paper: traces, $U(k)$, mmap/`O_DIRECT`, greedy 0.29/0.38 tok/s. Paper B is a speculation/engine *draft*. It is **not** for arXiv until a 30B `-v` smoke restores α and 235B 12-slot decode beats F10b `-ub 8` parity. Paper B is **not** this repository's PDF (https://github.com/Shriniwas410/capacity-law-serving/blob/main/paper/paper.pdf).

Abstract

Qwen3-235B-A22B stores 235B parameters and activates about 22B per token [Yang et al., 2025]. In 4-bit GGUF form the checkpoint is 133 GB—larger than a 32 GB desktop's RAM. We serve it on one Windows/WSL2 workstation (RTX 3070 8 GB idle; CPU-only llama.cpp) by streaming MoE experts from NVMe with `O_DIRECT`. Random 3.5/10/16 MB `O_DIRECT` QD8 reads on the ext4 VHDX measure 2.34–2.37 GB/s. CPU decode with an 8-slot expert cache is 0.29 tok/s; a 12-slot cache that still fits in 21 Gi of WSL RAM reaches 0.38 tok/s (peak RSS 16.70 GiB). Full CPU mmap dies allocating a 105 GB `CPU_REPACK` buffer.

We release 94-layer top-8 routing traces (4×2048 tokens). Mean unique experts over a 4-token window is $U(4) = 18.98$, so mean unique demand exceeds 16 slots (naive k=4 still fits 23% of 16-slot windows). Identical lag-1 expert *sets* occur 0.26% of the time; an 8-slot admission cap has mean admitted length T=1.003. Causal lag-2 prefetch recovers 0% of the LRU–OPT gap on these traces. Per-layer-mean oracle admission at 12 slots gives T=1.54 for k=2; the engine-relevant aggregate is the *minimum* over 94 layers, T=1.001 ($p(T{\geq}2)$=0.12%). This paper measures that geometry. Engine work that tries to speculate under the law is a separate manuscript.

1 Introduction

Mixture-of-Experts (MoE) layers route each token to a top-k subset of experts [Shazeer et al., 2017, Fedus et al., 2022]. That sparsity is why a 235B-total / 22B-active model [Yang et al., 2025] can run at all on a workstation, but "runs" is not "interactive." Offloading Mixtral-8×7B with an LRU expert cache has been shown to reach 2–3 tok/s on a T4, RTX 3060, or RTX 3080 Mobile [Eliseev and Mazur, 2023]; MoE-Infinity reports 3.1–16.7× latency reduction from expert-level traces on a

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commodity GPU [Xue et al., 2024]. Those results do not transfer to a 133 GB Qwen3-235B table on 21 Gi of WSL RAM.

This paper is a *measurement* study of that geometry. Direct I/O streaming of expert slabs is the substrate (llama.cpp PR#25294); we do not claim it as a novelty. The systems question we can answer now is how many *unique* experts a k-token window demands, and whether that union fits in a RAM-sized slot budget—a *capacity law*, not a bandwidth slogan. Speculative decoding [Leviathan et al., 2023] is why the union matters: verify is cheap only if $U(k) \leq s$. We do not report a speculation speedup here.

Contributions.

1. **A CPU tok/s ladder on one box** with provenance files: mmap impossible; 8-slot stream-direct 0.29 tok/s; 12-slot 0.38 tok/s at 16.70 GiB RSS.
2. **235B routing traces** of shape $(94, 2048, 8)$ on four domains, and the union law $U(k)$ that falsifies "just speculate with k=4."
3. **Negatives on the traces:** $U(4) \leq 16$ fails; causal prefetch recovery W=0; 8-slot Horizon-Spec mean admitted length T=1.003; engine-min T=1.001 at 12 slots k=2.

A 0.44 tok/s figure from Windows `llama-server -ngl 99 -ncmoe 94` (GPU non-MoE trunk, 2026-08-05) is a *different stack* and is not used below. The companion training-time locality paper [Suram, 2026] pre-registers 137M MoE routers with a locality loss: cache misses fall up to 60%, but every configuration fails a $\leq 1\%$ perplexity gate. We do not mix that result—or that paper's 0.44 tok/s warm 235B decode—into these CPU stream tables.

2 Background

MoE serving. At decode, each token's active experts must be present in a fast tier. EdgeMoE adapts expert bitwidth and keeps hot experts in an on-device buffer [Yi et al., 2023]. PowerInfer pins high-frequency neurons on GPU [Song et al., 2024]. Fiddler runs uncompressed Mixtral-8×7B by executing CPU-resident experts on the CPU [Kamahori et al., 2025]. FlexGen runs OPT-175B on a single 16 GB GPU at 1 tok/s generation *throughput* (effective batch size 144), not single-stream decode [Sheng et al., 2023]. Pre-gated MoE prefetches the next block's experts from the current block's gate [Hwang et al., 2023]. None of these papers report Qwen3-235B-A22B top-8 unions on a 12-slot, 21 Gi CPU stream cache.

Capacity law. Let s be the number of expert slots per MoE layer that fit in RAM, k the lookahead window, and $U(k)$ the number of unique experts in a k-token window at one layer. Greedy decode always loads $U(1)$ = top-8 = 8 experts per layer. A k-token step is one I/O generation only if $U(k) \leq s$. If an engine instead instantiates a worst-case graph of $k \times 8$ experts, extra waves pay compute and I/O even when the realized union is small. Admission-time $U_{\max}$=s would truncate a prefix so the union fits. On traces we simulate longest-prefix admission with *target* routing (an upper bound on sharing). That table is not tok/s.

3 Setup

Machine. Windows 11 + WSL2 Ubuntu, Intel i9-12900 (16C/24T), RTX 3070 8 GB idle, ~32 GB host RAM, `.wslconfig` 22 GB (WSL `free` ≈ 21 Gi), NVMe. llama.cpp branch `pr-25294`, git

Table 1: Unique experts $U(k)$ and lag-1 set reuse on Qwen3-235B-A22B top-8 traces. $U(4) \leq 16$ fails on every domain. Source: `measurements/uk_235b.json`.

domain	$U(4)$	$U(4)/32$	lag-1 reuse / chance	ratio
code	18.35	0.573	0.479 / 0.256	1.87×
general	18.87	0.590	0.463 / 0.203	2.27×
math	20.00	0.625	0.406 / 0.186	2.19×
medical	18.71	0.585	0.459 / 0.236	1.95×
mean	**18.98**	0.59	—	∼2.1×

`1248fd8fa`, `GGML_CUDA=OFF`. Model: `Qwen3-235B-A22B-Q4_K_M` (3-way split, 133 GB) on ext4. Expert cache `--moe-stream-direct --moe-stream-cache {8s,12s}`. Context 512, temperature 0, seed 1. Every headline number has a `config.json` / `verdict.json` under `measurements/` (Appendix A).

I/O. fio 3.39, 20 s, 30B GGUF on the same VHDX (`measurements/fio_summary.tsv`): mmap QD1 10 MB =0.064 GB/s; `pread` QD1 =1.377; `O_DIRECT` QD1 =1.856; `O_DIRECT` QD8 at 3.5/10/16 MB =2.368/2.339/2.354 GB/s.

30B check. Same binary, Qwen3-30B-A3B Q4_K_M: mmap 1.71 tok/s vs stream-direct 3.41 tok/s (1.99×). Direct I/O is not a no-op on this host.

4 The expert table versus the RAM slab

Qwen3-235B-A22B uses 94 MoE layers, 128 experts, top-8. At Q4_K_M an expert slab is 10.85 MB (`measurements/uk_umax_235b.json`). Greedy decode therefore demands $94 \times 8 \times 10.85\,\text{MB} = 7.97\,\text{GB}$ of expert bytes per token before cache hits. A 12-slot cache is $12 \times 94 \times 10.85\,\text{MB} = 11.96\,\text{GB}$ of expert RAM; measured peak RSS is 16.70 GiB including activations and the dense trunk (`measurements/ab235b_12s_verdict.json`). Sixteen slots would add ∼4 GiB on top of 16.7 GiB and were not attempted. Full mmap failed: `ggml_aligned_malloc` of 105 496 MB for `CPU_REPACK`.

Feasibility is *expert-table bytes divided by RAM slab*, not parameter count. DeepSeek-V3 is 671B total with 37B activated per token [DeepSeek-AI, 2024]; a 4-bit table for that geometry is larger than this 133 GB Qwen3 checkpoint. We do not claim it is a smooth fit here.

5 Routing traces and the union law

`llama-moe-trace` records top-8 ids with weights untouched (stream-direct, 8-slot cache, 2048 tokens × 4 domains, window 512). Shape $(94, 2048, 8)$. Provenance: `measurements/traces/` and `measurements/uk_235b.json`.

Union sharing is real ($U(4)/32 \approx 0.59$) but the union is larger than 16. Code $U(2) = 12.17$: mean unique experts over two tokens *exceeds* 12 slots. Naive k=2 fits 54% of 12-slot windows *per layer* (mean over layers), not a 4-token union and not the minimum over 94 layers.

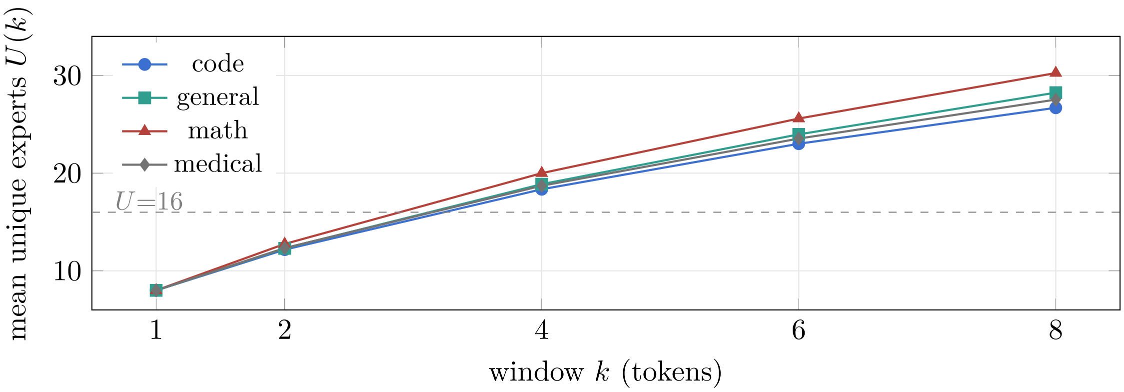

Figure 1: $U(k)$ grows through 16 by k=4 on every domain (`measurements/uk_235b.json`). Mean $U(4) = 18.98$ exceeds 16 slots; naive k=4 still fits 23% of 16-slot windows.

Table 2: Target-oracle HorizonSpec on 235B traces. Per-layer mean, then mean over four domains. Not engine min over layers, not tok/s. Source: `measurements/uk_umax_235b.json`.

cap	naive k=2 fit	HSpec k=2 T	naive k=4 fit	HSpec k=4 T
8 slots	0.26%	1.003	∼0%	1.003
12 slots	54.3%	1.54	1.2%	1.64
16 slots	100%	2.00	23.0%	2.83
24 slots	100%	2.00	94.6%	3.95

6 HorizonSpec is a slot count

Longest-prefix admission on these traces (target-oracle routing, an upper bound on union sharing) gives mean admitted length T. Consecutive tokens share an *identical* 8-expert set 0.26% of the time (`measurements/uk_umax_235b.json` pooled). With top-8 routing, $U_{\max}$=8 almost never admits a second token. Mean T at 8 slots is 1.003—greedy.

At 12 slots, naive k=2 fits 54% of windows and HorizonSpec mean T is 1.54 *per layer*. Taking min admitted length over 94 layers (`measurements/_min_layer_T.json`) gives pooled T=1.001, $p(T\geq 2)$=0.12%, ≈43 overflow layers per window. A global $U\leq 12$ gate therefore repeats T=1. Naive k=4 fits 1.2% of 12-slot windows: a four-token union still wants ∼19 unique experts. That is why "just set draft k=4" is the wrong lever on this router, independent of draft quality.

7 Greedy CPU decode

`llama-completion`, n=16, prompt "The capital of France is", `-b 1 -ub 1`. 8-slot page-cache 0.26 tok/s; 8-slot `O_DIRECT` 0.29 tok/s (`measurements/ab235b_verdict.json`). 12-slot `O_DIRECT` (colder first arm) **0.38 tok/s**, RSS 16.70 GiB (`measurements/ab235b_12s_verdict.json`). The same-day 8-slot rerun is warmer (0.30 tok/s) and is not averaged with the 12-slot number.

These rates are not interactive. They are the feasible CPU-stream ladder on this box. Lookahead (speculative decode, prefetch, admission) is constrained by $U(k)$ above; we do not table a speculation tok/s result in this manuscript.

Table 3: Demand misses per token (sum over 94 layers) at 16-expert cap. Lag-2 equals LRU. Source: `measurements/g2a_235b.json`.

domain	LRU	OPT W=8	lag-2	oracle-next	LRU hit
code	279.8	179.4	279.8	22.7	0.628
general	304.2	205.4	304.2	22.7	0.596
math	348.4	232.5	348.4	27.8	0.537
medical	293.6	193.2	293.6	25.1	0.610

8 Prefetch is not the lever

On the same traces, a per-layer cache with prefetch *charging a slot*, objective = demand misses (`measurements/g2a_235b.json`): at cap 16, lag-2 recovers 0% of the LRU→W8 gap; at caps 8 and 12 it *raises* demand misses. Frequency pinning is worse than LRU at cap 16 (code 307 vs 280 miss/tok). Non-causal oracle-next cuts demand misses ∼280→23. That ceiling is what a draft can supply only if a later engine can hold the union—admission-time $U_{\max}$, not lag-1 prefetch.

9 Limitations

All 235B decode numbers are one prompt length and one seed unless stated. Table 2 is per-layer-mean oracle routing, not engine tok/s; the min-over-layers figure is T=1.001 at 12 slots k=2 (`_min_layer_T.json`). We did not load 235B on the 8 GB GPU. The 0.44 tok/s GPU-split figure is excluded. Speculative-decode wall-clock and admission-engine experiments are not claims of this paper (see the scope box).

10 Related work

Shazeer et al. and Switch established sparse expert routing [Shazeer et al., 2017, Fedus et al., 2022]. Edge and offload serving include EdgeMoE [Yi et al., 2023], PowerInfer [Song et al., 2024], Mixtral offload [Eliseev and Mazur, 2023], MoE-Infinity [Xue et al., 2024], Fiddler [Kamahori et al., 2025], and pre-gated prefetch [Hwang et al., 2023]. FlexGen studies offload scheduling [Sheng et al., 2023]. Speculative decoding verifies draft tokens in parallel [Leviathan et al., 2023, Cai et al., 2024, Li et al., 2024]. Concurrent MoE-speculative papers name a verify *union* on GPU/HBM or by dropping experts [McDanel et al., 2026, Liang et al., 2026, Xie et al., 2026, Pan et al., 2026]. Those levers are not a 12-slot CPU stream. Training-time router locality on 137M models is a companion negative result [Suram, 2026] and is not this paper's tok/s ladder. We contribute the 235B top-8 union law on a RAM slab that does *not* hold the table.

11 Conclusion

On this box, 235B Q4_K_M serving is a 12-slot, ≈0.3–0.4 tok/s CPU stream. The traces say why lookahead is hard: mean $U(4)$ = 18.98, 8-slot admitted length T=1.003, and engine-min T=1.001 at 12 slots k=2. Causal prefetch does not close the LRU–OPT gap. The publishable claims are the traces, the union law, the mmap/`O_DIRECT` ladder, and those negatives—not a speculation speedup. We post this as a measurement preprint. It is not a TMLR resubmission of the companion training-time locality paper [Suram, 2026].

Reproducibility. Scripts, verdicts, and traces live under `measurements/` (https://github.com/Shriniwas410/capacity-law-serving). llama.cpp `1248fd8fa`. CPU-only; GPU idle. Appendix A lists the file for each headline number.

Use of AI assistance. The author used an AI coding assistant to help implement the experimental harness and to draft and edit the manuscript. All experimental design decisions, the runs, and the interpretation are the author's, who takes full responsibility for the content and has verified every reported number against the listed artifacts.

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A Provenance

Table 4: Headline numbers and the files that contain them. Do not overlay rows that differ in binary, prompt, or cache size.

claim	value	file under `measurements/`
O_DIRECT QD8	2.34–2.37 GB/s	`fio_summary.tsv`
30B mmap / direct	1.71 / 3.41 tok/s	`ab30b/`
$U(4)$ mean	18.98	`uk_235b.json`
identical lag-1 sets	0.26%	`uk_umax_235b.json`
HSpec per-layer-mean T @ 12s k=2	1.54	`uk_umax_235b.json`
engine min T @ 12s k=2	1.001	`_min_layer_T.json`
mmap OOM	105 496 MB	`ab235b/mmap_repack_oom.stderr`
8s O_DIRECT greedy	0.29 tok/s	`ab235b_verdict.json`
12s O_DIRECT greedy	0.38 tok/s, 16.70 GiB	`ab235b_12s_verdict.json`
lag-2 @ cap 16	0% of LRU–W8 gap	`g2a_235b.json`